

The Quadratic Formula

For a quadratic equation in standard form

$$ax^2 + bx + c = 0, \quad a \neq 0,$$

the solutions are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Before substituting

Write the equation in standard form and identify a , b , and c with their signs. Use parentheses when substituting negative values.

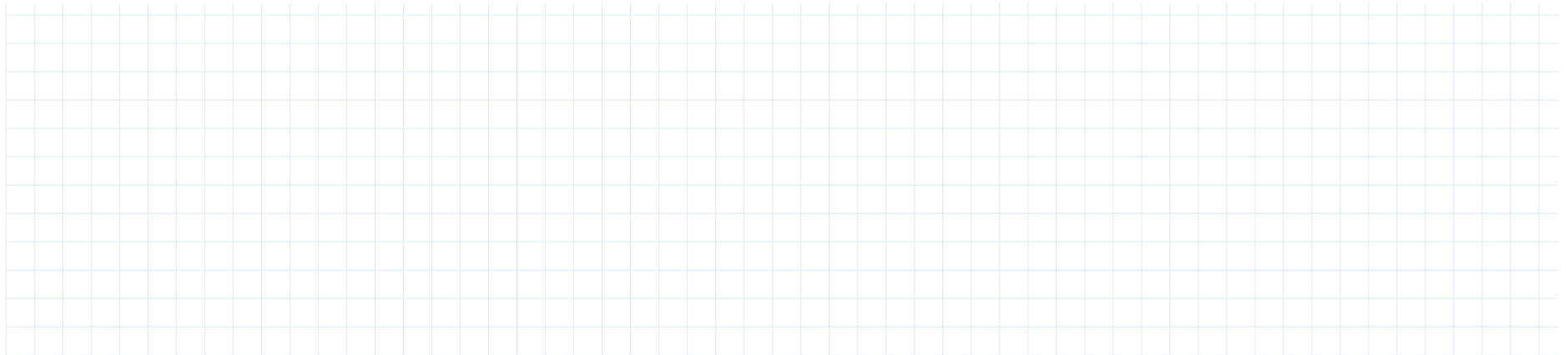
For $6x^2 - 5x - 4 = 0$, identify a , b , and c .

Use the Quadratic Formula

Solve

$$6x^2 - 5x - 4 = 0.$$

1. Identify a , b , and c .
2. Substitute into the formula without simplifying.
3. Simplify the discriminant and radical.
4. Reduce each solution.



The Discriminant

The expression under the radical is the **discriminant**:

$$D = b^2 - 4ac.$$

Value of D	Number of real solutions	Type of solutions
$D > 0$, perfect square	two	rational
$D > 0$, not a perfect square	two	irrational
$D = 0$	one repeated solution	rational
$D < 0$	no real solutions	two complex conjugates

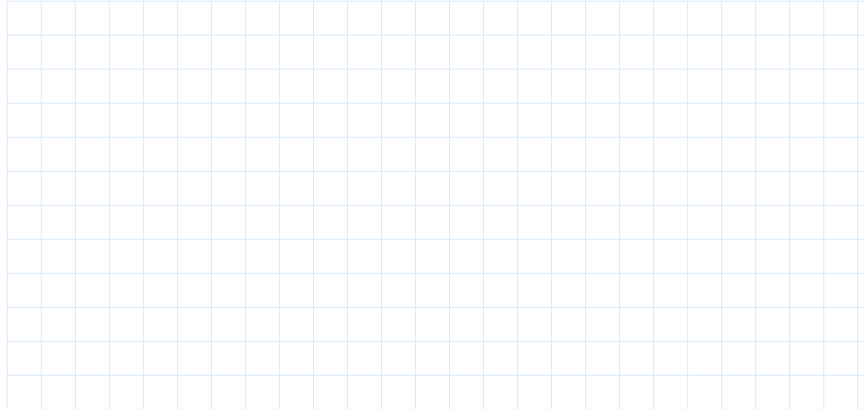
The discriminant predicts the number of x -intercepts before we solve.

Classify Before Solving

Find the discriminant. State the number and type of solutions; do not solve the equations.

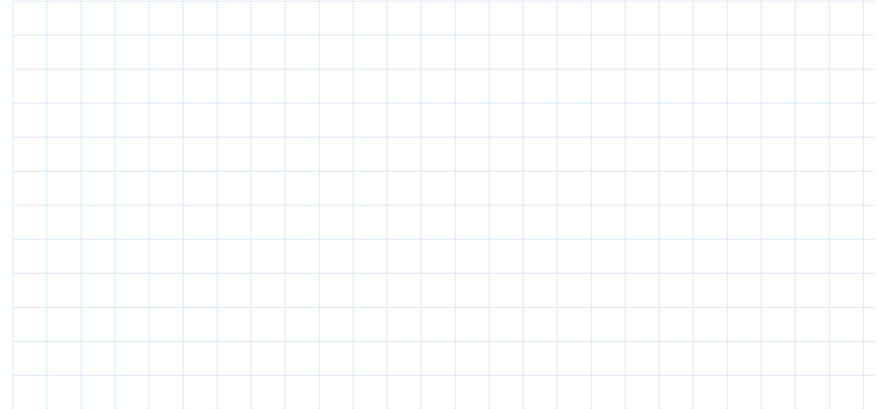
1.

$$6x^2 - x - 15 = 0$$



2.

$$3x^2 - 4x - 5 = 0$$

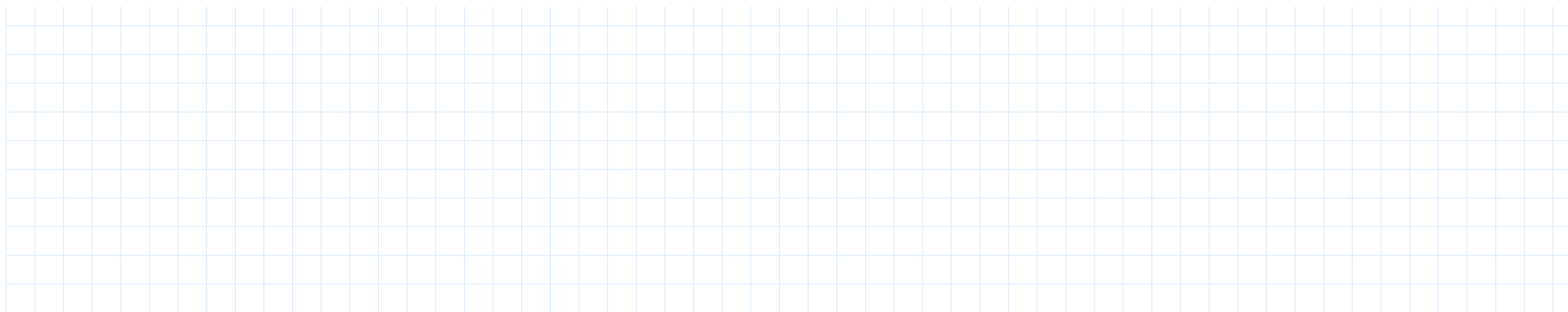


Intersections of Two Functions

To find the intersection points of $y = f(x)$ and $y = g(x)$:

1. Solve $f(x) = g(x)$ for all possible x -values.
2. Substitute each x -value into either original function.
3. Report each intersection as an ordered pair (x, y) .

Example. Let $f(x) = x^2 + 5x - 3$ and $g(x) = 2x + 1$. Find all intersection points.



Functions in Quadratic Form

Some higher-degree equations become quadratic after a substitution.

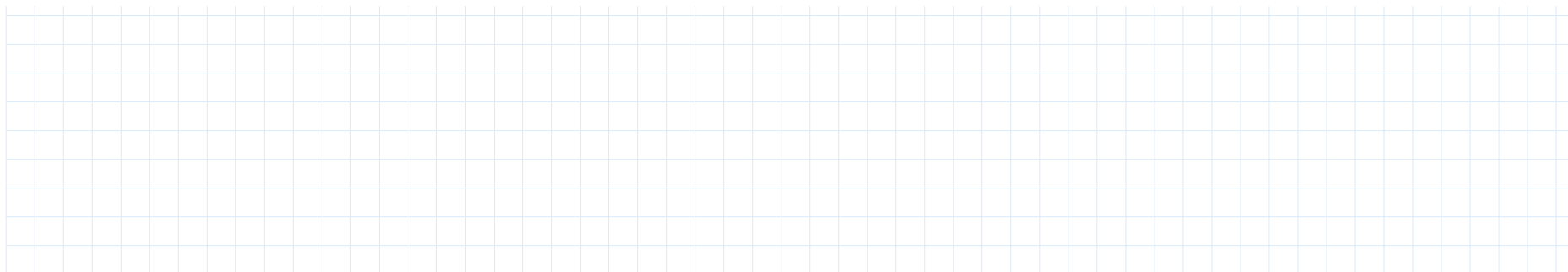
Substitution pattern

If the equation contains $[u(x)]^2$ and $u(x)$, set

$$v = u(x).$$

Solve the resulting quadratic equation in v , then back-substitute to solve for x .

For $x^4 + x^2 - 12 = 0$, choose a substitution and rewrite the equation in quadratic form.

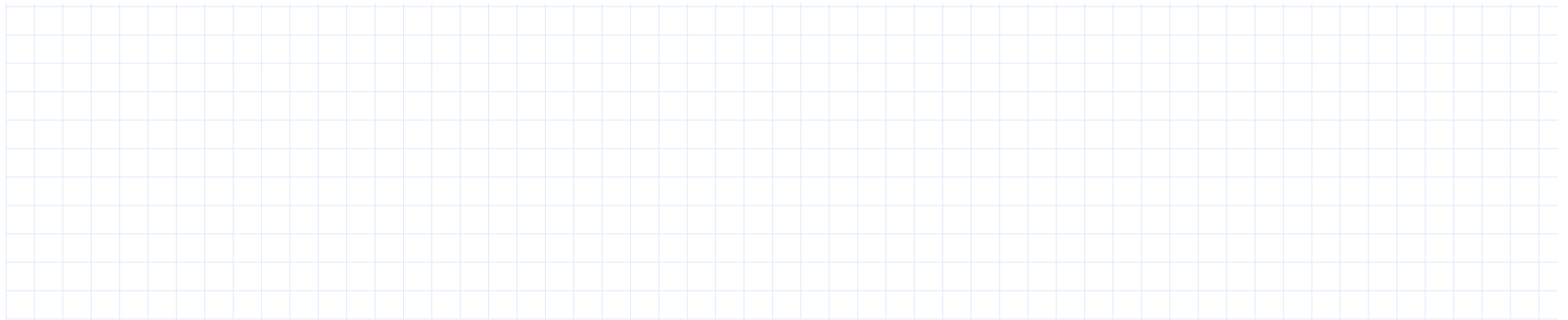


Solve by Substitution

Solve completely:

$$x^4 + x^2 - 12 = 0.$$

1. Let $v = x^2$.
2. Solve the quadratic equation in v .
3. Back-substitute each value of v .
4. Keep track of real and nonreal solutions.



Mixed Quadratic Practice

Choose an efficient method and show complete work.

1. Solve $4x^2 - 8x + 1 = 0$.
2. Use the discriminant to classify the solutions of $2x^2 + 3x + 7 = 0$.
3. Find the intersections of $y = x^2 - 1$ and $y = x + 1$.
4. Solve $(x + 2)^2 + 11(x + 2) - 12 = 0$ using substitution.

